Division B Rules Manual

Division B (JSS 1- 3)

WELCOME TO THE 2026 SCIENCE OLYMPIAD!

This Rules Manual will help you prepare to compete in the State and National Tournaments holding across Nigeria. All users of this manual are subject to the Terms of Use Agreement. To compete, users must first join the Science Olympiad program in their state and become registered members.

See our website for info on Membership, Policies and Terms of Use at www.sciolyng.org

Division C (SSS 1 - 3) Membership Rules

A team may have up to twelve (12) members. A maximum of six (6) SSS 3 students is permitted on a Division C team.

Division B (JSS 1 - 3) Membership Rules

A team may have up to twelve (12) members. A maximum of six (6) JSS 3 students is permitted on a Division B team. Minimum four (4) JSS 2 students, and two (2) JSS 1 students.

Students Below Grade Level Designations

Science Olympiad encourages students to participate in the Division that matches current Science Olympiad grade level designations. However, to support the inclusion of students who wish to participate in Science Olympiad, schools with grade levels lower than those stated in a Division are permitted to invite members below the grade level designations. Participation is limited to age-appropriate events (as determined by a coach, principal or tournament director) and prohibited where safety is a concern (such as the use of chemicals). See Team Qualifications for more information.

Find more Science Olympiad team information under the Policies section of the national website: Code of Ethics & Rules, Scoring Guidelines, Home & Virtual Schools, Small Schools, All Stars, Copyrights and Use, Lasers, Building Policy, Eye Protection, Significant Figures and Wristband Procedures.

Exploring the World of Science

SCIENCE OLYMPIAD
DIVISION B RULES MANUAL

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- Please read the General Rules on the next page - they apply to all events. Note: all changes are in **bold**.
- Please visit the official Science Olympiad web site: www.soinc.org for team registration, further Information, Team Size Requirements, Clarifications/Rules Changes, FAQs, New Store Items, news, tips, resources, and other valuable information.

Exploring the World of Science

GENERAL RULES, CODE OF ETHICS, AND SPIRIT OF THE PROBLEM

The goal of competition is to give one's best effort while displaying honesty, integrity, and good sportsmanship. Everyone is expected to display courtesy and respect - see Science Olympiad Pledges. Teams are expected to make an honest effort to follow the rules and the spirit of the problem (not interpret the rules so they have an unfair advantage). Failure by a participant, coach, or guest to abide by these codes, accepted safety procedures, or rules below, may result in an assessment of penalty points or, in rare cases, disqualification by the tournament director from the event, the tournament, or future tournaments.

1. Actions and items (e.g., tools, notes, resources, supplies, electronics, etc.) are permitted, unless they are explicitly excluded in the rules, are unsafe, or violate the spirit of the problem.
2. While competing in an event, participants may not leave without the event supervisor's approval and must not receive any external assistance. All electronic devices capable of external communication **as well as calculator applications on multipurpose devices (e.g., laptop, phone, tablet) are not permitted unless expressly permitted in the event rule or by an event supervisor**. Cell phones, if not permitted, must be turned off. At the discretion of the event supervisor, participants may be required to place their cell phones in a designated location.
3. Participants, coaches and other adults are responsible for ensuring that any applicable school or Science Olympiad policy, law, or regulation is not broken. All Science Olympiad content such as policies, requirements, clarifications/changes and FAQs on www.sciolyng.org must be treated as if it were included in the printed rules.
4. All pre-built devices presented for judging must be constructed, impounded, and operated by one or more of the 12 current team members unless stated otherwise in the rules. If a device has been removed from the event area, appeals related to that device will not be considered.
5. Officials are encouraged to apply the least restrictive penalty for rules infractions - see examples in the Scoring Guidelines. Event supervisors must provide prompt notification of any penalty, disqualification or tier ranking.
6. State and national tournament directors must notify teams of any site-dependent rule or other rule modification with as much notice as possible, ideally at least 30 days prior to the tournament.

Exploring the World of Science

ANATOMY & PHYSIOLOGY

1. **DESCRIPTION:** Participants will be assessed on their understanding of the anatomy and physiology for the human Cardiovascular, Lymphatic, and Excretory systems.

   A TEAM OF UP TO: 2 **APPROXIMATE TIME:** 50 Minutes

2. **EVENT PARAMETERS:**
   Each team may bring one blank sheet of paper for rough work or information analysis.

3. **THE COMPETITION:**
   Participants will complete a written test limited to the following topics.
   a. **CARDIOVASCULAR SYSTEM:**
      i. Anatomy and **physiology** of the cardiovascular system
      ii. The Heart - chambers and valves of the heart, electrical stimulation of myocardial tissue, pacemaker tissue, interpreting ECG (EKG) readings **on strips**
      iii. Blood Vessels – **structure and function of** arteries, arterioles, veins, venules, capillaries, **including the functionality of Startling's forces in the capillaries**
      iv. Blood - plasma, hematocrit, red blood cells, oxygen transport, hemoglobin **and cooperative binding of oxygen**, platelets and blood clotting, regulation of blood plasma volume and acidity, blood type & basic genetics of ABO, Rh, blood types
      v. Measurement of the pulse rate and blood pressure **with appropriate instrumentation**
      vi. Calculations include systolic and diastolic pressure, mean arterial pressure, stroke volume & cardiac output
      vii. Effects of exercise, smoking, alcohol, caffeine, and drugs on the cardiovascular system
      viii. Understand disorders: Congestive Heart Failure, Atrial Fibrillation, Myocardial Infarction, Atherosclerosis, Bradycardia, and Tachycardia

   b. **LYMPHATIC SYSTEM:**
      i. Anatomy and **physiology** of the lymphatic system
      ii. **Similarities and differences between Primary, Secondary, and Tertiary lymphoid tissues**
      iii. **General Lymphatic structures -** lymph nodes, lymph ducts, lymphatic capillaries, tissue fluid
      iv. **Structure and function of the Thymus**
      v. **Structure and function of the Spleen**
      vi. **Understand disorders: Lymphedema, Hodgkin lymphoma, non-Hodgkin lymphoma, Lymphadenopathy**

   c. **EXCRETORY SYSTEM:**
      i. Anatomy and **physiology** of the excretory system
      ii. Basic anatomy of the urinary system including kidneys, ureters, bladder, and urethra
      iii. Structure and function of the nephron
      iv. Formation of urine, **Gross Filtration Rate** calculation, tubular secretion, and tubular absorption
      v. Understand disorders: **Kidney stones, Urinary Tract Infections, Glomerulonephritis, Renal failure**, **Incontinence**

4. **SCORING:**
   a. High score wins.
   b. Selected questions will be used to break ties.

SCIENCE OLYMPIAD
Exploring the World of Science

BATTERY BUGGY

1. **DESCRIPTION:** Teams will construct a vehicle that uses electrical energy as its sole means of propulsion to travel a specified distance and stop as close as possible to a **Target** Point.

   A TEAM OF UP TO: 2 **IMPOUND:** Yes **EVENT TIME:** 8 minutes

2. **EVENT PARAMETERS:**
   a. Participants must bring **and impound** a vehicle, batteries, additional/spare parts, and a Practice Log.
   b. Participants may bring but **need not impound** tools, measuring devices, and a stand-alone calculator of any type.

3. **CONSTRUCTION PARAMETERS:**
   a. Participants will design a vehicle to travel between 9 and 12 meters as quickly as possible and come to a complete stop. **The team may choose to attempt a Bonus described later in these rules**.
   b. Electrical energy used by the vehicle for any purpose, including propulsion, must be stored in a maximum of 8 (eight) AA 1.2 to 1.5-volt common, commercially available batteries, individually labeled by the manufacturer. Rechargeable batteries are allowed. Batteries must not be installed until immediately prior to the run. All sources of energy must be in a location as to be available for inspection by the Event Supervisor. Battery holders are permissible, as long as the labels on the individual cells can be inspected during impound.
   c. **The vehicle must be constructed so that when it comes to a complete stop the batteries are no longer powering the motor.**
   d. **Any battery containing lithium or lead acid is not permitted**. Teams using these batteries will not be permitted to run and will receive only participation points.
   e. All energy for propulsion must be electric and come from the batteries. Any non-propulsive functions (e.g., a braking system) may be powered by either electric or non-electric storage devices. If electrical, the voltage must come from the same batteries being used for propulsion.
   f. Participants may purchase or make components (e.g., motors, gearboxes, bodies, and chassis). Electrical components are limited to batteries, wires, motors, switches, resistors, potentiometers, and mechanical relays. **Electric/electronic tools of any sort, except a stand-alone calculator, are prohibited**.
   g. Wheels and/or treads in their entirety must fit in a 30.0 cm wide x 60.0 cm long space during the entire run. **Axles and other parts of the vehicle may extend beyond these parameters**.
   h. **An approximately ¼" round** wooden dowel must be attached to the front end of the vehicle. The dowel must be approximately perpendicular to the floor. It must be the leading part of the vehicle at all times. **The dowel attachment device may not extend more than 0.5 cm beyond the front of the dowel**.
      i. The dowel must extend at least 20.0 cm above the floor.
      ii. The dowel must also extend to within 1.0 cm of the track's surface so that its front bottom edge will be the vehicle's Measurement Point for distance measurements.
   i. Only the wheels and/or treads of the vehicle are allowed to contact the floor at any time during run(s).
   j. Participants must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy found on www.sciolyng.org.

4. **PRACTICE LOG:**
   a. **Teams must record the vehicle distance from the Target Point, run time, wheel setting, and Bonus can distance, if used, of at least 10 practice runs while varying at least one vehicle parameter.**
   b. Logs will be impounded **and returned to participants when they are called to compete**.

5. **THE COMPETITION:**
   a. The exact Target Distance (in 50.0 cm intervals for regional, 10.0 cm intervals for state and 1.0 cm intervals for national tournaments) will be chosen by the Event Supervisor and announced after the impound period. All teams will have the same Target Distance. **Participants must inform the Event Supervisor if they will attempt the Bonus prior to each run**.
   b. Only the participants and the Event Supervisors are allowed in the impound and event areas. Once participants enter the event area, they must not leave or receive outside assistance, materials, or communication.
   c. Participants will be given an Event Time of 8 minutes to perform the following actions and start up to 2 runs. If the second run has started before the 8-minute period has elapsed, it will be allowed to run to completion. The Event Time will not include time used by the Event Supervisor for measuring.

Participants may not use AC outlet power during their 8 minutes.

i. Participants may adjust their vehicle before each run (e.g., change its speed, distance, directional control, change batteries, or make changes from impounded parts).

ii. Participants may use their own measuring devices to verify the track dimensions during their allotted time. They may not verify the distance by rolling the vehicle on or adjacent to the track surface between the Start and Target Points. They may not roll the vehicle on the floor of the event track the day of the event without tournament permission. If permitted, only participants may be present.

iii. Participants may clean the track during their 8 minutes, but the track must remain dry at all times. No wet and/or tacky substances may be applied to the track, wheels, or treads.

iv. Participants must place the tip of the vehicle's Measurement Point on the Start Point and align the vehicle. Sighting and aligning devices placed on the track are permitted but must be removed before the run. Vehicle-mounted sighting and aligning devices may be removed at the participants' discretion prior to each run.

v. Participants must start the vehicle using any part of an unsharpened #2 pencil with an unused eraser, supplied by the Event Supervisor, **in a motion approximately perpendicular to the floor**, to actuate some sort of switch. They may not touch the vehicle to start it, hold it while actuating the switch, or "push" the vehicle to get it started.

vi. The entire vehicle, including batteries, must move forward together. A run occurs if the vehicle moves after the switch is actuated.

vii. If the vehicle does not move upon actuation, it will not count as a run and the team may request to set up for another run but will not be given additional time.

d. Once the vehicle begins a run, the participants must wait until called by the Event Supervisor to retrieve it. **Timing is paused for all measurements**. The 8-minute time resumes once participants pick up their vehicle or begin to make measurements.

e. Competition Violations would include participants following the vehicle down the track or the vehicle passing the 0.5 m Line but stopping before the 8.50 m Line.

f. A Failed Run occurs if the time and/or distance cannot be measured for a run (e.g., the vehicle starts before the Event Supervisor is ready, the participants pick up the vehicle before it is measured, the vehicle's Measurement Point doesn't reach the 0.50 m Line, the vehicle runs backward at the start, the vehicle does not stop using its own braking system).

g. **Teams may earn the Bonus by having their vehicle navigate between two cans located on the Bonus Line of the track. Prior to each run the participants place the inner can at a distance of their choice between 0.0 cm and 100.0 cm from the outside can on the Bonus Line. The Event Supervisor will then record the inner distance between the two cans along the Bonus Line. All parts of the vehicle must travel between the two cans to earn the Bonus. A vehicle moving either of the cans will not receive the Bonus.**

h. The Event Supervisor will review with the team the data and penalties recorded on their scoresheet.

i. Teams who wish to file an appeal must leave their vehicle with the Event Supervisor.

6. **<u>THE TRACK</u>:**

a. The track will be on a smooth, level, and hard surface. At the Event Supervisor's discretion, more than one track may be used. Teams will be given the option to choose which track they will use. **Both runs will be made on the same track**.

b. The Event Supervisor will use approximately 5.0 cm by 2.5 cm (2 in. x 1 in.) pieces of tape to mark the Start and Target Points with the Start and Target Point marked in the center of each piece of tape. The distance between the Start and Target Points will be measured to within 0.1 cm of the Target Distance. **The timing lines are marked with pieces of 2.5 cm wide tape at least 2.0 m long, 0.50 m and 8.50 m from the Start Point, centered on and perpendicular to an imaginary center line. The edges of the tapes closer to the Start Point defines these lines. A Bonus Line is marked with tape halfway between the Start Point and Target Point extending perpendicular from an imaginary center line 1.0 m to the left when facing the Target Point.**

c. **The Event Supervisor will use two weighted cans with diameters 7.0 - 8.0 cm, height ≧ 10.5 cm positioned standing upright with their centers on the Bonus Line to create an obstacle for the teams. The outer can is placed by the Event Supervisor so that its rightmost edge is 1.0 m from the imaginary center line. The cans will be removed if the team does not want to attempt the Bonus for a specific run.**

BATTERY BUGGY (CONT.)

7. **SCORING:**
   a. Low score wins. The Final Score is the better of the two Run Scores; negative scores are possible.
   b. **Run Score for each run = Run Time** + 2x Distance Score + **Bonus + Penalties**.
   c. Run Time starts when the dowel on the vehicle reaches 0.50 m and ends when it either completely stops or passes 8.50 m. The Run Time is recorded in seconds to the precision of the timing device used.
   d. The Distance Score is a point to point measurement of the distance from the Measurement Point to the Target Point measured to the nearest 0.1 cm.
   e. **Bonus = -0.5 x (110 - distance between the cans to the nearest 0.1 cm).**
   f. Teams with incomplete Practice Logs will incur a Penalty of 250 points. Teams without impounded Practice Logs will incur a Penalty of 500 Points. Practice Log Penalties do not affect Tier placement.
   g. Tiers:
      i. Tier 1: A run with no violations.
      ii. Tier 2: A run with any competition violations.
      iii. Tier 3: A run with any construction violations.
      iv. Tier 4: A vehicle not impounded during the impound period.
   h. Ties will be broken the following categories in the listed order:
      i. the lower Bonus Distance
      ii. the lower Distance Score
      iii. the lower Time Score
      iv. the lower score of the other run
   i. Teams who cannot complete a run within their 8 minutes or have 2 Failed Runs will be given participation points.
   j. Scoring Example: The run took 8.53 seconds, came to rest 10.4 cm from the Target Point with a Bonus distance of 50.0 cm, and the vehicle incurred no Penalties.

Run Time	8.53 seconds	=	8.53	pts.
Distance Score	2 x 10.4 cm	=	20.8	pts.
Bonus	[-0.5 x (110-50.0)]	=	-30.0	pts.
+ Penalties	0	=	0.0	pts.
Run Score			-0.67	pts.

Exploring the World of Science

DENSITY LAB

1. **DESCRIPTION:** Participants compete in activities and answer questions about mass, density, number density, area density, concentration, pressure, and buoyancy.

   A TEAM OF UP TO: 2 **EYE PROTECTION:** B **APPROXIMATE TIME:** 50 minutes

2. **EVENT PARAMETERS:**
   a. Each team may also bring writing utensils, and two stand-alone calculators of any type for use during any part of the event.
   b. Event supervisors will provide any material and measurement devices required for the hands-on tasks. Teams will not use their own measurement devices.

3. **THE COMPETITION:**
   Part I: Written Test
   a. The written test consisting of multiple choice, true-false, completion, or calculation questions/problems will assess the team's knowledge of mass, density, number density, area density, concentration, temperature, pressure, and buoyancy.
   b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
   c. The test will consist of at least one question from each of the following areas:
      i. Density of solids, liquids, and gases
      ii. Determination of concentrations limited to: mass/mass, mass/volume, volume/volume percentages, parts per million (ppm), and parts per billion (ppb)
      iii. Behavior of gases according to the gas laws: Ideal Gas, Boyle's, Charles', Gay-Lussac's, and Avogadro's
      iv. Archimedes' Principle

   Part II: Hands-On Tasks
   a. The hands-on portion of the competition will consist of at least one task at a station(s) for the teams to complete.
   b. **Tasks, or stations, will relate to the above content and may include but are not limited to**:
      i. For a given container of gas, measure its volume and mass, and calculate the mass density.
      ii. Given a small bag of Skittles, determine the number density of the green Skittles in the bag.
      iii. Given a helium balloon and a balance determine the mass that the balloon could theoretically lift.
      iv. Determine the depth to which a solid object will sink when placed in water.
      v. Determine the density of a material at different temperatures (e.g., air or water).

4. **SCORING:**
   a. High score wins.
   b. The written portion of the competition will account for 50-75% of each team's score. No single question will count for more than 10% of the total points possible on the written test.
   c. The hands-on portion of the competition will account for the remaining 25-50% of each teams score.
   d. Points will be awarded for correct answers, measurements, calculations, and data analysis. Supervisors are encouraged to provide a standard form for competitors to show measurements/calculations.
   e. Ties will be broken using pre-selected task(s)/question(s) that will be noted on the written test.

Exploring the World of Science

DISEASE DETECTIVES

1. **DESCRIPTION:** Participants will use their investigative skills in the scientific study of disease, injury, health, and disability in populations or groups of people.

   A TEAM OF UP TO: 2 **APPROXIMATE TIME:** 50 minutes

2. **EVENT PARAMETERS:**
   Each team may bring one blank sheet of paper.

3. **THE COMPETITION:**
   a. **This event has been reorganized into three parts with each part counting approximately equally towards a team's final score.**

   Part I: Background & Surveillance
   a. Understand the Clinical Approach (health of individuals) and Public Health Approach (health of populations)
   b. Understand the roles of epidemiology in public health and the steps in solving health problems
   c. Understand the Natural History and Spectrum of Disease and the Chain of Infection
   d. Understand the basic epidemiological and public health terms (e.g., outbreak, epidemic, pandemic, surveillance, risk, vector)
   e. Understand the role of Surveillance in identifying health problems, the 5 step Process for Surveillance and the types of Surveillance

   Part II: Outbreak Investigation
   a. Analyze an actual or hypothetical outbreak
   b. Understand the Types of Epidemiological Studies – Experimental and Observational
   c. Be able to identify the Steps in an Outbreak Investigation
   d. Identify the problem using person, place, and time triad – formulate case definition
   e. Interpret epi curves, line listings, cluster maps, and subdivided tables
   f. Generate hypotheses using agent, host, and environment triad
   g. Recognize various fundamental study designs and which is appropriate for this outbreak
   h. Evaluate the data by calculating and comparing simple rates and proportions as attack rate, relative risk, odds-ratio, and explaining their meaning

   Part III: Patterns, Control, and Prevention
   a. Identify patterns, trends of epidemiologic data in charts, tables, and graphs
   b. Using given data, calculate disease risk and frequencies as a ratio, proportion, incidence proportion (attack rate), incidence rate, prevalence, or mortality rate
   c. Understand the Strategies of Disease Control
   d. Understand Strategies for Prevention - the Scope and Levels of Prevention

4. **SCORING:**
   a. High score wins. Selected questions may be used as tiebreakers.
   b. Points will be assigned to the various questions and problems. Both the nature of the questions and scoring will emphasize an understanding that is broad and basic rather than detailed and advanced.
   c. Depending on the problem, scoring may be based on a combination of answers, including graphs/charts, explanations, analysis, calculations, and closed-ended responses to specific questions.
   d. Points will be awarded for both quality and accuracy of answers, the quality of supporting reasoning, and the use of proper scientific methods.

Exploring the World of Science

DYNAMIC PLANET

1. **DESCRIPTION:** Students will use process skills to complete tasks related to glaciers, glaciation, and long-term climate change.

   A TEAM OF UP TO: 2 **APPROXIMATE TIME:** 50 minutes

2. **EVENT PARAMETERS:**
   a. Each team may bring four blank sheets of paper for information analysis.
   b. Each team may bring two **stand-alone non-programmable, non-graphing** calculators.

3. **THE COMPETITION:**
   Participants will be presented with one or more tasks, many requiring the use of process skills (e.g., observing, classifying, measuring, inferring, predicting, communicating, and using number relationships) from the following topics:
   a. Glacier formation: Properties of ice, ice crystal structure, and formation of glacial ice from snow & firn
   b. Glacial mass-balance and flow: ablation and accumulation zones, equilibrium line, influence of bed (wet or dry, bare rock, and sediment), and relation of flow to elevation and gradient
   c. Glacier/ice sheet types and forms: valley/**alpine (cirque, hanging, piedmont)**, ice sheet/**continental including** ice stream, ice shelf, ice rise, ice cap, ice tongue, & the geographic distribution **of these features**
   d. Glacial features: crevasses, ogives, icefalls, what they are, & what they indicate about flow and melt
   e. Formation of landscape features:
      i. Erosional - cirque, tor, U-shaped valley, hanging valleys, aretes, horns, **striations & grooves, and Rôche moutonnée**
      ii. Depositional – moraines **(end/terminal, recessional, lateral, medial, ground),** kettles, kames, drumlins, eskers, and **erratics**
   f. Glacial hydrology: Surface melt, surface lakes, moulins, drainage and subglacial lakes, & Jökulhlaups
   g. Global connections of glaciation:
      i. Atmosphere - greenhouse gases, insolation, and aerosols
      ii. Oceans - sea level change and ice sheet variation
      iii. Lithosphere - Isostatic effects on Earth's crust
   h. History of ice on Earth:
      i. Neoproterozoic snowball Earth
      ii. Late Paleozoic ice ages
      iii. Eocene Oligocene Transition and the impact of opening oceanic seaways
      iv. Pleistocene onset of Northern Hemisphere glaciation
   i. Ice cores as archives of past environments including gases, aerosols, and stable isotope composition
   j. Sedimentary sequences produced in glacial environments in the marine and terrestrial realms
   k. The Laurentide Ice Sheet retreat & melting history; impact on river drainage; and oceanic circulation
   l. Modeling rates and size of ice sheet changes (e.g., marine ice sheet instability, ice shelf buttressing)
   m. Methods of studying glaciers & what they tell you: Altimetry, radar, Landsat, seismology, and gravity
   n. Recent records of cryospheric change: (e.g., Larsen B, Kilimanjaro, Amundsen Sea Embayment)

4. **SAMPLE QUESTIONS/TASKS:**
   a. Analyze and interpret glacial features on a USGS topographic map or satellite image.
   b. Analyze a geologic map of glacial deposits to determine the sequence of events over the course of several episodes of advance and retreat.
   c. Interpret oxygen isotope data from a marine sediment core to identify changes in sea level caused by global ice volume changes.
   d. Apply glaciological principles to predict where one might find meteorites in ice fields.

5. **SCORING:**
   a. High score wins. Points will be awarded for the quality and accuracy of responses.
   b. Ties will be broken by the accuracy and/or quality of answers to pre-selected questions.

Exploring the World of Science

MYSTERY ARCHITECTURE

1. **DESCRIPTION:** At the beginning of the event, teams will be given a bag of building materials and instructions for designing and building a device that can be tested.

   A TEAM OF UP TO: 2 **APPROXIMATE TIME:** 50 minutes

2. **EVENT PARAMETERS:**
   a. Each team may bring 1 pair of scissors, **1 flat standard 30 cm (12") ruler**, and 1 pair of pliers.
   b. No other materials, tools, notes, or resources are permitted.

3. **THE COMPETITION:**
   a. Each team will be given a bag containing the same materials and instructions as to the type of device to be constructed. The students will not know the task until they begin the competition.
   b. Examples of materials that may be provided include, but are not limited to: paper cups, drinking straws, paper clips, string, tape, paper, thumbtacks, and craft sticks. Only those materials contained in the bag may be used to build the device. The bag and instructions must not be used. No other materials or adhesives may be part of the finished device.
   c. The devices to be built are limited to an **elevated** bridge, cantilever, **arch** or **tunnel**. If a cantilever is to be built, the event supervisor will supply the fulcrum and provide a counterbalance. **If a tunnel is to be built, the event supervisor will specify the internal dimension**.
   d. The instructions must identify a Primary Dimension, a Secondary Dimension, whether the device must support a load, and the required duration of load support.
   e. Unless specifically stated in the instructions, devices must be freestanding and must not be attached to a tabletop, floor, ceiling, or other support.
   f. If the device must support a load, a separate identical load of the same dimensions and weight as used for testing will be provided to each team. When finished building, the load must be removed from the device. The event supervisor will direct the participants when to place the official load in/on the device.
   g. Only participants and the event supervisor are allowed in the event area. Once in the event area, they must not leave or receive outside assistance, materials, or communication.
   h. The supervisor will review with the team the data being recorded on their score sheet.

4. **SAMPLE TASKS & PRIMARY DIMENSIONS:**
   a. For an **elevated** bridge, the Primary Dimension could be the **measurement** between the closest inside supports **plus the height from the base to the lowest bridge support**. If the bridge fails to support the load, the Primary Dimension will be measured from the point of contact to the farther inside support **plus a height score of zero**.
   b. For a cantilever, the Primary Dimension could be measured:
      i. with no load, from the fulcrum to the end of the cantilever,
      ii. with a load, from the fulcrum to the closest point of contact or attachment of the load.
   c. **For an arch, the Primary Dimension could be measured:**
      i. **with no load, from the base to the highest point of the arch**
      ii. **with a load, from the base to the highest point of the load**
   d. For a **tunnel**, the Primary Dimension could be **the measurement of the longest continuously enclosed portion of the tunnel**.

5. **SCORING:**
   a. Highest or lowest score wins depending on instructions.
   b. The Primary and Secondary Dimensions will be measured in cm to the nearest 0.1 cm by the event supervisor. Devices requiring a load will be measured both prior to and after placement of the load **once the duration time concludes**, if successfully held.
   c. Devices without load requirements will be ranked in order of Primary Dimensions as per instructions.
   d. Devices with load requirements will be ranked as follows:
      i. Tier 1: Devices which support the load will be ranked in order of Primary Dimensions after the placement of the load.
      ii. Tier 2: Devices where the load, or its underlying material, contact the table or the event supervisor is unable to measure the height due to movement of the load will be ranked by Primary Dimensions as measured before the placement of the load.
   e. The Secondary Dimension will be used as a Tie Breaker if necessary.

THERMODYNAMICS

1. **DESCRIPTION:** Teams must construct an insulating device prior to the tournament that is designed to retain heat and complete a written test on thermodynamic concepts.

   A TEAM OF UP TO: 2 **EYE PROTECTION:** C

   IMPOUND: Yes **APPROXIMATE TIME:** 50 Minutes

2. **EVENT PARAMETERS:**
   a. Each team may bring one three-ring binder of any size containing information in any form and from any source attached using the available rings. **Participants may remove pages during the event**.
   b. Each team may also bring tools, supplies, writing utensils, and two **stand-alone** calculators of any type for use during any part of the event. These items need not be impounded.
   c. Each team must impound: their insulating device; **an** unaltered, glass or plastic, standard (height ~1.4 times the diameter) 250 mL beaker; **a device diagram and** copies of graphs and/or tables for scoring.
   d. Event supervisors will supply the hot water, devices for transferring measured volumes of water, **cotton balls**, and thermometers or probes (recommended).
   e. Prior to competition, teams must calibrate devices by preparing graphs/tables showing the relationship **between water temperature and testing parameters**. A labeled device picture/diagram should be included.
      i. Any number of graphs and/or data tables may be submitted but the team must indicate up to four to be used for the Chart Score, otherwise the first four provided are scored.
      ii. Graphs and/or tables may be computer generated or drawn by hand on graph paper. **Each data series counts as a separate graph**. A template is available at www.soinc.org.
      iii. Teams are encouraged to have a duplicate set to use, as those submitted may not be returned.
   f. Participants must wear eye protection during Part I. Teams without proper eye protection must be immediately informed and given a chance to obtain eye protection if time allows.
   g. Participants must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy found on www.sciolyng.org.

3. **CONSTRUCTION PARAMETERS:**
   a. Devices may be constructed of and contain anything **except** the following materials/components: asbestos; mineral wool; fiberglass insulation; **commercially available thermoses/coolers/vacuum sealed devices**.
   b. The device must fit within a:
      i. 20.0 x 20.0 x 20.0 cm cube for Division B
      ii. 15.0 x 15.0 x 15.0 cm cube for Division C
   c. Within the device, participants must be able to insert and remove a beaker that they supply (see 2.c.).
   d. The device must also allow putting a thermometer/probe into the beaker via a hole ≧ **0.50 inches** in diameter all the way through directly above the beaker. The hole's top surface must be < 12 cm above the inside bottom beaker surface. **The hole's bottom surface may be inside the beaker but must not contact the water. Teams may plug the hole with a single cotton ball provided by the supervisor**.
   e. Devices will be inspected to ensure that there are no energy sources (e.g., electric components, battery powered heaters, chemical reactions, etc.) to help keep the water warm. At the event supervisor's discretion, teams must disassemble devices after testing in order to verify the construction materials.
   f. All parts of the device must not be significantly different from room temperature at impound.

4. **THE COMPETITION:**

   Part I: Device Testing
   a. At the start of each competition block, the supervisor will announce the volume of water (**Regionals: 100 mL; States: 75 - 125 mL, 25 mL increments; Nationals: 75 - 125 mL, 5 mL increments**) and the cooling time (Div. B: **25.0** mins; Div. C: **20.0-30.0** mins, **1-minute increments**). These parameters will be the same for all teams.
   b. The event supervisor will announce the temperature of the source water bath (60 - **75** ^{o}C) and the current room temperature. **Supervisors will do their best to keep this the same for all teams but must announce the actual values (in case of minor fluctuations) at the start of each competition block**.
   c. At the start of the competition block, teams will be given 5 minutes to set up or modify their devices and use their graphs and/or tables to begin temperature prediction calculations. Devices that do not meet the construction specs will not be allowed to be tested until brought into specification.

d. **The supervisor, using his/her own measuring device, will dispense the volume of water into each team's beaker. A team may elect to install a beaker in a device prior to this but must leave sufficient access to the beaker**. Teams may secure/close access panels with fastening materials after **receiving water, but must do so in a manner to not delay dispensing to other teams**. Supervisors must record the time each team receives water and the room and source water temperature when dispensed.
e. Teams will use their graphs and/or tables to calculate the temperature of the water in their beaker at the end of the cooling time. **After receiving water, teams will be given at least 3, but no more than 5 minutes to make their final predictions. During this time, teams may use their own thermometers to measure the starting water temperature in their beaker, but after this time must remove them**.
f. At the end of the cooling period, the supervisor will record the ending time and the temperature in the beaker to the best precision of the available instrument. Supervisors may leave thermometers/probes in the devices for the entire cooling period but will announce if they will do so before impound. Otherwise, they will insert a thermometer/probe into **the beaker in the device**, wait at least 20 seconds, and record the resulting temperature. Multiple thermometers/probes may be used at the supervisor's discretion.
g. The supervisor will review with the team the Part I data recorded on their scoresheet.
h. Teams filing an appeal regarding Part I must leave their device in the competition area.

Part II: Written Test

a. Teams will take a test on thermodynamics during the remaining time after all devices receive water.
b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
c. Teams will be given a minimum of 20 minutes to complete a written test consisting of multiple choice, true-false, completion, or calculation questions/problems.
d. The test will consist of at least **three** questions from each of the following areas:
   i. The history of thermodynamics
   ii. **Definition of temperature**, temperature scales and conversions, definitions of heat units
   iii. **Phases of matter, phase transitions, phase diagrams, latent heat, ideal gas law**
   iv. **Kinds of heat transfer, thermal conductivity, heat capacity, specific heat**
   v. Thermodynamic laws and processes (e.g., Carnot cycle and efficiency, adiabatic, isothermal)
   vi. **Division C only: Radiant exitance, entropy, enthalpy**

5. **SCORING:**
   a. High score wins. All scoring calculations are to be done in degrees Celsius (°C).
   b. The Final Score = TS + CS + HS + PS; a scoring spreadsheet is available at www.soinc.org.
   c. Test Score (TS) = (Part II score / Highest Part II score for all teams) x 45 points
      i. Chart Score (CS) = max of 10 points
      ii. Heat Score (HS) = **20 x (lowest k of all teams) / k**, **where k is from Newton's law of cooling: k = - (1 / cooling time) x ln((start water temp - room temp) / (final water temp - room temp))**
      iii. Prediction Score (PS) = **25 - 2.5*abs(prediction - final temp)**. The minimum PS possible is 0 points.
   d. One of the submitted graphs and/or tables, selected by the event supervisor, must be scored as follows for the Chart Score. Partial credit may be given.
      i. 2 points for including data spanning at least one variable range listed in **4.Part I.a.**
      ii. 2 points for including at least 10 data points **in each data series**
      iii. 2 points for proper labeling (e.g. title, team name, units)
      iv. 0.5 points for each graph or table turned in (up to 2 points total as long as they are not the same)
      v. 2 points for including a labeled device picture or diagram
   e. If a team violates any COMPETITION rules, their **PS score** will be multiplied by 0.9 and **their k will be multiplied by 1.1** when calculating the scores.
   f. If any CONSTRUCTION violation(s) are corrected during Part I, or if the team misses impound, their **PS** will be multiplied by 0.7 and their **k will be multiplied by 1.4** when calculating the scores.
   g. Teams disqualified for unsafe operation or do not having a conforming insulating device at the start of Part I receive zero points for their **HS and PS** scores. Teams will be allowed to compete in Part II.
   h. Tie Breakers will be applied in the following order: i. Best TS, ii. Best PS, iii. Best HS.

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CALCULATOR GUIDE

The following document was prepared to offer some guidance to teams as they select calculators for use in different Science Olympiad events. By no means are the calculators listed here inclusive of all possible calculators; instead they are offered as common examples. **The decisions of the event supervisors will be final.**

Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators can be used in the following events: Anatomy & Physiology, Battery Buggy, Density Lab, Disease Detectives, Dynamic Planet, Thermodynamics,

Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators are the most basic type of calculators and often look like the one shown to the right. These calculators are limited to the four basic mathematics functions and sometimes square roots. These calculators can often be found at dollar stores.

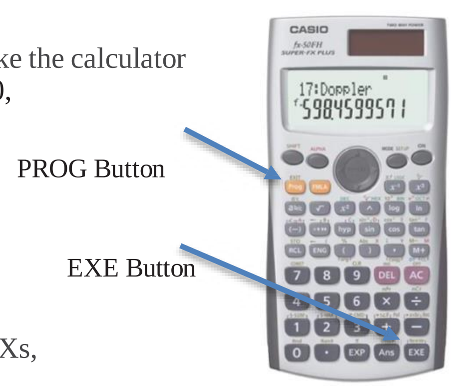
Stand-alone non-programmable, non-graphing calculators, in addition to the above listed calculators, can be used in the following events: Anatomy & Physiology, Dynamic Planet, Disease Detectives.

Stand-alone non-programmable, non-graphing calculators look like the calculator to the right or simpler. There are hundreds of calculators in this category but some common examples include: CASIO FX-260, Sharp EL-501, and TI-30X.

Stand-alone, programmable, graphing calculators and **stand-alone non-graphing, programmable calculators,** in addition to the above listed calculators, can be used in the following events: Battery Buggy, Density Lab, Thermodynamics.

Stand-alone, programmable, graphing calculators often look like the calculator shown on the left. Some examples are: Casio 975 0/9850/9860, HP 40/50/PRIME, and TI 83/84/89/NSPIRE/VOYAGE.

Stand-alone non-graphing, programmable calculators are another type of calculator that can be used in the above listed events. To identify these calculators, look for the presence of the 'EXE' button, the 'Prog' button, or a 'file' button. Examples include but are not limited to: Casio Super FXs, numerous older Casio models, and HP 35S. A calculator of this type with the buttons labeled is shown to the right.



Calculator applications on multipurpose devices (e.g., laptop, phone, tablet, watch) are not allowed unless expressly permitted in the event rule.

EYE PROTECTION

This resource was created to help teams comply with the Science Olympiad Policy on Eye Protection.

Participant/Coach Responsibilities: Participants are responsible for providing their own protective eyewear. Science Olympiad is unable to determine the degree of hazard presented by equipment, materials and devices brought by the teams. Coaches must ensure the eye protection participants bring is adequate for the hazard. All protective eyewear must bear the manufacturer's mark. At a tournament, teams without adequate eye protection will be given a chance to obtain eye protection if their assigned time permits. If required by the event, participants will not be allowed to compete without adequate eye protection. This is **non-negotiable**.

Corresponding Standards: Protective eyewear used in Science Olympiad must be manufactured to meet the Standards applicable at its time of manufacture. Competitors, coaches and event supervisors are not required to acquire a copy of the standard. The information in this document is sufficient to comply with current standards. Water is not a hazardous liquid and its use does not require protective eyewear unless it is under pressure or substances that create a hazard are added.

Compliant Eyewear Categories: If an event requires eye protection, the rules will identify one of these three categories. Compliance is simple as ABC:

CATEGORY A

- Description: Non-impact protection. They provide basic particle protection only
- Corresponding ANSI designation/required marking: Z87
- Examples: Safety glasses; Safety spectacles with side shields; and Particle protection goggles (these seal tightly to the face completely around the eyes and have direct vents around the sides, consisting of several small holes or a screen that can be seen through in a straight line)

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+
- Example: High impact safety goggles

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you are able to see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)
- Example: Indirect vent chemical/splash protection goggles

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved in lieu of the primary eye protection devices below regardless of the type of vents they have.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87 mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for all three Categories A, B & C
2. VisorGogs do not seal completely to the face, but are acceptable as indirect vent chemical/splash protection goggles

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